

Advanced Level Mathematics Practice Paper

Calculus: Curve Sketching & Applications of Differentiation

20 Structured Exam-Style Questions modelled on A/L Past Paper Standards

Instructions: Answer all questions. Each question follows the standard structural format of Advanced Level structured essay questions, testing asymptotic behaviour, first derivatives, monotonicity, turning points, second derivatives, points of inflection, and graphical representation.

1. Let $f(x) = \frac{x^2 + x + p}{(x - 1)^2}$ for $x \in \mathbb{R} - \{1\}$, where $p \in \mathbb{R}$. It is given that the graph of $y = f(x)$ intersects its horizontal asymptote at a point with the x -coordinate $-\frac{1}{3}$.

(a) Show that $p = 2$.

(b) Show that $f'(x) = -\frac{3x + 5}{(x - 1)^3}$ for $x \in \mathbb{R} - \{1\}$.

(c) Hence, find the intervals on which $f(x)$ is increasing and the intervals on which $f(x)$ is decreasing.

(d) Also, find the coordinates of the turning point of the graph of $y = f(x)$.

(e) It is given that $f''(x) = \frac{6(x + 3)}{(x - 1)^4}$ for $x \in \mathbb{R} - \{1\}$. Show that the graph of $y = f(x)$ has a point of inflection at $(-3, \frac{1}{2})$.

(f) Sketch the graph of $y = f(x)$ indicating the turning point, the asymptotes, and the point of inflection.

2. Let $f(x) = \frac{x^2 - 2x + p}{(x - 2)^2}$ for $x \in \mathbb{R} - \{2\}$, where $p \in \mathbb{R}$. It is given that the graph of $y = f(x)$ intersects its horizontal asymptote at $x = 0$.

(a) Show that $p = 3$.

(b) Show that $f'(x) = \frac{2(x - 1)}{(x - 2)^3}$ for $x \in \mathbb{R} - \{2\}$.

(c) Find the intervals of increase and decrease of $f(x)$, and the coordinates of any turning points.

(d) Given that $f''(x) = -\frac{2(2x + 1)}{(x - 2)^4}$, determine whether there is any point of inflection.

(e) Sketch the graph of $y = f(x)$ showing all key features.

3. Let $f(x) = \frac{2x^2 + x + p}{(x + 1)^2}$ for $x \in \mathbb{R} - \{-1\}$, where $p \in \mathbb{R}$. The graph of $y = f(x)$ intersects its horizontal asymptote at $x = 1$.

(a) Show that $p = 4$.

(b) Show that $f'(x) = -\frac{3(x - 3)}{(x + 1)^3}$.

(c) Find the intervals where $f(x)$ is increasing/decreasing and locate the turning point.

(d) Given $f''(x) = \frac{6(x - 5)}{(x + 1)^4}$, find the point of inflection and sketch the curve.

4. Let $f(x) = \frac{x^2 - 4x + p}{(x - 3)^2}$ for $x \in \mathbb{R} - \{3\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 1$.

(a) Show that $p = 7$.

(b) Show that $f'(x) = -\frac{2(x - 5)}{(x - 3)^3}$.

(c) Determine the monotonicity and turning points of $f(x)$.

(d) Given $f''(x) = \frac{4(x - 7)}{(x - 3)^4}$, find the point of inflection and sketch the graph.

5. Let $f(x) = \frac{x^2 + 3x + p}{(x + 2)^2}$ for $x \in \mathbb{R} - \{-2\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 0$.

(a) Show that $p = 2$.

(b) Show that $f'(x) = -\frac{x + 1}{(x + 2)^3}$.

(c) Find the intervals of increase/decrease and the turning point coordinates.

(d) Given $f''(x) = \frac{2x - 1}{(x + 2)^4}$, analyze concavity and sketch the curve.

6. Let $f(x) = \frac{2x^2 - 3x + p}{(x - 1)^2}$ for $x \in \mathbb{R} - \{1\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 2$.

(a) Show that $p = 1$.

(b) Show that $f'(x) = \frac{x - 1}{(x - 1)^3}$ (simplify appropriately) / $\frac{5}{(x - 1)^3}$ – let's verify: $f(x) = \frac{2x^2 - 3x + 1}{(x - 1)^2} = \frac{(2x - 1)(x - 1)}{(x - 1)^2} = \frac{2x - 1}{x - 1}$ for $x \neq 1$.

(c) Find $f'(x)$ and determine its sign for all $x \neq 1$.

(d) Discuss asymptotes, turning points, and sketch the graph.

7. Let $f(x) = \frac{x^2 + 4x + p}{(x + 2)^2}$ for $x \in \mathbb{R} - \{-2\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 1$.

(a) Show that $p = 5$.

(b) Show that $f'(x) = -\frac{4}{(x + 2)^3}$.

(c) Determine intervals of increase/decrease, turning points (if any), and sketch the graph.

8. Let $f(x) = \frac{x^2 - x + p}{(x - 1)^2}$ for $x \in \mathbb{R} - \{1\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 2$.

(a) Show that $p = 2$.

(b) Show that $f'(x) = -\frac{1}{(x-1)^3}$.

(c) Find the intervals of monotonicity and sketch the graph.

9. Let $f(x) = \frac{3x^2 + 2x + p}{(x+1)^2}$ for $x \in \mathbb{R} - \{-1\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 0$.

(a) Show that $p = 1$.

(b) Show that $f'(x) = -\frac{4(x)}{\dots}$ let's write: $f'(x) = -\frac{4}{(x+1)^3}$.

(c) Find turning points, monotonicity, and sketch the graph.

10. Let $f(x) = \frac{x^2 - 5x + p}{(x-3)^2}$ for $x \in \mathbb{R} - \{3\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 1$.

(a) Show that $p = 8$.

(b) Show that $f'(x) = -\frac{x-7}{(x-3)^3}$.

(c) Find turning points, intervals of increase/decrease, and sketch the graph.

11. Let $f(x) = \frac{x^2 + 2x + p}{(x - 2)^2}$ for $x \in \mathbb{R} - \{2\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = -1$.

(a) Show that $p = 0$.

(b) Show that $f'(x) = -\frac{6(x)}{\dots}$ i.e., $f'(x) = -\frac{2(3x + 2)}{(x - 2)^3}$.

(c) Find turning points, intervals of increase/decrease, point of inflection given $f''(x)$, and sketch.

12. Let $f(x) = \frac{2x^2 - x + p}{(x - 1)^2}$ for $x \in \mathbb{R} - \{1\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 0$.

(a) Show that $p = 1$.

(b) Show that $f'(x) = \frac{3}{(x - 1)^3}$.

(c) Analyze monotonicity, asymptotes, and sketch the graph.

13. Let $f(x) = \frac{x^2 + 5x + p}{(x + 3)^2}$ for $x \in \mathbb{R} - \{-3\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 1$.

(a) Show that $p = 8$.

(b) Show that $f'(x) = -\frac{x - 1}{(x + 3)^3}$.

(c) Find turning points, intervals of increase/decrease, and sketch the curve.

14. Let $f(x) = \frac{x^2 - 3x + p}{(x - 2)^2}$ for $x \in \mathbb{R} - \{2\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 1$.

(a) Show that $p = 4$.

(b) Show that $f'(x) = -\frac{x - 5}{(x - 2)^3}$.

(c) Find turning points, intervals of increase/decrease, and sketch the graph.

15. Let $f(x) = \frac{3x^2 + x + p}{(x + 1)^2}$ for $x \in \mathbb{R} - \{-1\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 2$.

(a) Show that $p = 2$.

(b) Show that $f'(x) = -\frac{x - 3}{(x + 1)^3}$.

(c) Find turning points, monotonicity, and sketch the graph.

16. Let $f(x) = \frac{x^2 + 6x + p}{(x + 3)^2}$ for $x \in \mathbb{R} - \{-3\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 0$.

(a) Show that $p = 9$.

(b) Show that $f'(x) = -\frac{3}{(x+3)^3}$.

(c) Discuss monotonicity, asymptotes, and sketch the graph.

17. Let $f(x) = \frac{2x^2 + 3x + p}{(x-1)^2}$ for $x \in \mathbb{R} - \{1\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = -2$.

(a) Show that $p = 1$.

(b) Show that $f'(x) = -\frac{7x+5}{(x-1)^3}$.

(c) Find turning points, intervals of increase/decrease, and sketch the curve.

18. Let $f(x) = \frac{x^2 - 2x + p}{(x-1)^2}$ for $x \in \mathbb{R} - \{1\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 3$.

(a) Show that $p = 3$.

(b) Show that $f'(x) = -\frac{2}{(x-1)^3}$.

(c) Discuss monotonicity, asymptotes, and sketch the graph.

19. Let $f(x) = \frac{x^2 + 4x + p}{(x + 2)^2}$ for $x \in \mathbb{R} - \{-2\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 3$.

(a) Show that $p = 7$.

(b) Show that $f'(x) = -\frac{2(x + 1)}{(x + 2)^3}$.

(c) Find turning points, intervals of increase/decrease, and sketch the graph.

20. Let $f(x) = \frac{3x^2 - 4x + p}{(x - 2)^2}$ for $x \in \mathbb{R} - \{2\}$, where $p \in \mathbb{R}$. The graph intersects its horizontal asymptote at $x = 1$.

(a) Show that $p = 1$.

(b) Show that $f'(x) = -\frac{2(x - 5)}{(x - 2)^3}$.

(c) Find turning points, intervals of increase/decrease, point of inflection, and sketch the complete graph of $y = f(x)$ indicating all key geometric features.

— End of Practice Paper —

Advanced Level Combined Mathematics Practice Paper

Calculus: Comprehensive Differentiation & Curve Sketching (Questions 21–40)

Extension Set: 20 Additional Distinct Exam-Style Questions covering Advanced Parametric, Implicit, Trigonometric Series, and Optimization

Instructions: Answer all questions. This extension paper features 20 brand-new unique logic structures matching the rigorous standards of the Sri Lankan G.C.E. Advanced Level Combined Mathematics examination.

21. Let $f(x) = \frac{2x}{x^2 - 4}$ for $x \in \mathbb{R} - \{-2, 2\}$.

- (a) Find the vertical asymptotes and the horizontal asymptote of the graph of $y = f(x)$.
- (b) Show that $f'(x) = \frac{-2(x^2 + 4)}{(x^2 - 4)^2}$.
- (c) Determine whether $f(x)$ has any turning points or points of inflection, and state the intervals of monotonicity.
- (d) Sketch the graph of $y = f(x)$ showing all key features.

22. Let $f(x) = x^2 e^{-x}$ for $x \in \mathbb{R}$.

- (a) Find $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$.
- (b) Show that $f'(x) = x(2 - x)e^{-x}$.
- (c) Find the coordinates and nature of all turning points of $f(x)$.
- (d) Given $f''(x) = (x^2 - 4x + 2)e^{-x}$, find the coordinates of the points of inflection and sketch the graph.

23. Let $f(x) = \arcsin\left(\frac{2x}{1+x^2}\right)$ for $x \in [-1, 1]$.

- (a) By using the substitution $x = \tan \theta$, simplify $f(x)$ for $-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}$.
- (b) Hence, find $f'(x)$ directly for $-1 < x < 1$ and verify your result.
- (c) Discuss the differentiability of $f(x)$ at $x = 1$ and $x = -1$.

24. Let $f(x) = \ln(x^2 + 1) - x \arctan x$ for $x \in \mathbb{R}$.

- (a) Find $f'(x)$ and simplify your expression as much as possible.
- (b) Show that $f'(1) = 1 - \frac{\pi}{4}$.
- (c) Determine the critical points of $f(x)$ and find the equation of the tangent to the curve at $x = 0$.

25. A curve is given parametrically by $x = ct$ and $y = \frac{c}{t}$ for $t \in \mathbb{R} - \{0\}$, where $c > 0$ is a constant.

- (a) Find $\frac{dy}{dx}$ in terms of t .
- (b) Find the equation of the tangent to the curve at the point $P\left(ct_1, \frac{c}{t_1}\right)$.
- (c) Show that the area of the triangle formed by this tangent and the coordinate axes is independent of t_1 .

26. A curve is defined implicitly by $\sin x + \cos y = \frac{1}{2}$ for $x, y \in (0, \pi)$.

- (a) Find $\frac{dy}{dx}$ in terms of x and y .
- (b) Find the coordinates of the point on the curve where $x = \frac{\pi}{6}$ and y is in the given interval.
- (c) Find the equation of the normal to the curve at this point.

27. A right circular cylinder is inscribed in a sphere of fixed radius $R = 6$ cm.

- (a) Express the volume V of the cylinder as a function of its radius r .
- (b) Using differentiation, find the radius r and height h of the cylinder that maximize its volume.
- (c) Calculate the maximum volume of the inscribed cylinder.

28. Let $f(x) = \frac{x}{(x-1)^2}$ for $x \in \mathbb{R} - \{1\}$.

- (a) Find the vertical and horizontal asymptotes of $y = f(x)$.
- (b) Show that $f'(x) = -\frac{x+1}{(x-1)^3}$.
- (c) Find the intervals of increase and decrease, and the coordinates of the turning point.
- (d) Sketch the graph of $y = f(x)$.

29. Let $f(x) = \cos^3 x \sin 2x$ for $x \in \mathbb{R}$.

- (a) Express $f(x)$ in terms of $\sin x$ and $\cos x$ powers.
- (b) Find $f'(x)$ and factorize it fully.
- (c) Find all critical points of $f(x)$ in the interval $[0, \pi]$.

30. Sand is pouring from a chute onto a conical pile at a constant rate of $12 \text{ cm}^3\text{s}^{-1}$. The height of the pile is always equal to half of the diameter of its base.

- (a) Express the volume V of the conical pile as a function of its height h only.
- (b) Find the rate at which the height of the pile is increasing when the height is 4 cm.
- (c) Find the rate at which the circular base area is increasing at that same instant.

31. Let $f(x) = \frac{x^2 - 1}{x^3}$ for $x \in \mathbb{R} - \{0\}$.

- (a) Find the intercepts of the graph with the coordinate axes and determine any vertical or horizontal asymptotes.
- (b) Show that $f'(x) = \frac{3 - x^2}{x^4}$.
- (c) Find the local extrema of $f(x)$ and specify their nature.
- (d) Sketch the graph of $y = f(x)$.

32. Let $f(x) = \ln(x^2 - 4x + k)$ for a real constant k .

- (a) Given that the domain of $f(x)$ is $\mathbb{R} - [1, 3]$ (or adjust so roots are 1 and 3), find the value of k .
- (b) For this value of k , find $f'(x)$ and simplify it.
- (c) Find the turning point of $f(x)$ and determine its nature.
- (d) Find the vertical asymptotes of the graph.

33. A curve is given parametrically by $x = a \cos^3 \theta$ and $y = a \sin^3 \theta$ for $\theta \in [0, 2\pi]$, where $a > 0$.

- (a) Find $\frac{dy}{dx}$ in terms of θ .
- (b) Find the equation of the tangent to the curve at the point where $\theta = \frac{\pi}{4}$.
- (c) Show that the sum of the intercepts of any tangent to this curve on the coordinate axes is constant and equal to a .

34. Let a curve be defined by $\ln(xy) + x + y = 4$ for $x, y > 0$.

- (a) Find $\frac{dy}{dx}$ implicitly in terms of x and y .
- (b) Verify that the point $(1, e)$ or a suitable point lies on the curve (or test $x = 1, y = e^3$, let's use $xy = e^{4-x-y}$; at $x = 1, y = 1 \implies \ln(1) + 1 + 1 = 2 \neq 4$; let's set proper point: point $(1, e^2)$ or rewrite equation as $\ln x + \ln y + x + y = 4$. If $x = 1, y = 1 \implies 0 + 0 + 1 + 1 = 2$. Let's use $xye^{x+y} = e^2$ or $\ln x + \ln y + x + y = 2$). Let's formulate: curve given by $\ln(xy) + x + y = 2$. Verify $(1, 1)$ lies on the curve.
- (c) Find the equation of the tangent at $(1, 1)$.

35. A right circular cone is inscribed in a sphere of fixed radius R .

- (a) Express the volume V of the cone in terms of the sphere's radius R and the distance x from the center of the sphere to the base of the cone.
- (b) Using differentiation, prove that the volume of the cone is maximized when $x = \frac{R}{3}$. Find the maximum volume of the inscribed cone in terms of R .

36. Let $f(x) = \frac{x^2 - 3x + 2}{x - 3}$ for $x \in \mathbb{R} - \{3\}$.

- (a) Find the vertical asymptote and the oblique asymptote of the graph of $y = f(x)$.
- (b) Show that $f'(x) = \frac{x^2 - 6x + 7}{(x - 3)^2}$.
- (c) Find the coordinates of the turning points and determine their nature.
- (d) Sketch the graph of $y = f(x)$.

37. Let $f(x) = e^{2x} \cos 3x$ for $x \in [0, \pi]$.

- (a) Show that $f'(x) = e^{2x}(2 \cos 3x - 3 \sin 3x)$.
- (b) Express $f'(x)$ in the form $Re^{2x} \cos(3x + \alpha)$, stating the exact value of R and α .
- (c) Find the critical points of $f(x)$ in the given interval and determine the absolute extrema.

38. Let $y = (\arcsin x)^2$ for $x \in [-1, 1]$.

- (a) Show that $\sqrt{1 - x^2} \frac{dy}{dx} = 2 \arcsin x$.
- (b) By differentiating the above relation again with respect to x , show that $(1 - x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 2$.
- (c) Find the value of $\frac{d^2y}{dx^2}$ at $x = 0$.

39. A rectangle is bounded by the x -axis and the parabola $y = 9 - x^2$ in the upper half-plane.

- (a) Express the area A of the rectangle as a function of the position x of the positive vertex on the x -axis ($0 < x < 3$).
- (b) Using differentiation, find the dimensions of the rectangle that yield the maximum area.
- (c) Verify that this yields a maximum and state the maximum area.

40. Let $f(x) = \frac{x}{(x^2 + 1)(x + 2)}$ for $x \in \mathbb{R} - \{-2\}$.

- (a) Find the vertical asymptote and the horizontal asymptote of the graph of $y = f(x)$.
- (b) Find $f'(x)$ and express it as a single simplified rational fraction.
- (c) Determine the critical points of $f(x)$ and discuss its monotonicity.
- (d) Sketch the complete graph of $y = f(x)$ incorporating all asymptotic and turning point behaviors.